Python Engineering - Class 06 Theory Guide

Dictionaries and Sets

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Association vs position

Lists answer 'what is at position i?'. Dictionaries answer 'what value belongs to this named key?'.

Relationship / equation

```
index -> value vs key -> value
```

Connection to code

Engineering configuration becomes easier to read when g, dt and h0 are named explicitly.

2. Configuration contract

A simulation expects a minimum set of parameters. Set operations express this requirement compactly.

Relationship / equation

```
missing = required - provided
```

Connection to code

If missing is empty, the structural configuration requirement is satisfied.

3. Uniqueness

A set stores unique elements, which makes it useful for membership and comparing groups of names.

Relationship / equation

```
{'g','dt','g'} -> {'g','dt'}
```

Connection to code

The goal is not ordered storage but logical set operations.

4. Results as dictionaries

Returning labelled outputs reduces ambiguity when a function produces many engineering quantities.

Relationship / equation

```
results['max_altitude']
```

Connection to code

This becomes more maintainable than remembering tuple positions as the number of outputs grows.

Manual validation method

For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

Named configuration/results prepare the course for scalable simulation interfaces and structured outputs.

Final summary

```
MODEL / CONCEPT
      |
      v
ALGORITHM
      |
      v
PYTHON IMPLEMENTATION
      |
      v
REFERENCE CHECK
      |
      v
ENGINEERING INTERPRETATION
```

> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.